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CUET UG

Physics (322)

Current Electricity

Official Syllabus Unit 2 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 2)	Key Exam Focus & Skills
Electric Current & Microscopic Conduction	Electric current, flow of electric charges in a metallic conductor, drift velocity, mobility and their relation with electric current.	Drift equation $I = neAv_d$, relaxation time τ, mobility $\mu = \frac{v_d}{E}$, current density vector (J).
Ohm's Law & Resistivity	Ohm's law, electrical resistivity and conductivity, temperature dependence of resistance, V-I characteristics (linear and non-linear).	Microscopic Ohm's law $(J) = \sigma(E)$, resistivity $\rho = \frac{m}{ne^2\tau}$, temperature coefficient α, non-ohmic devices (diode, thyristor).
Electrical Energy & Power	Electrical energy and power, heating effects of current, rated vs actual power, maximum power transfer theorem.	Joule's law $P = I^2R = \frac{V^2}{R}$, bulb brightness in series vs parallel, condition for max power from non-ideal sources $(R_L = r)$.
EMF, Internal Resistance & Cells	Internal resistance of a cell, potential difference and emf of a cell, combination of cells in series and in parallel.	Discharging $(V = E - Ir)$ vs charging $(V = E + Ir)$, series (nE, nr), parallel $(E, \frac{r}{m})$, mixed grouping, unequal parallel cells.
Kirchhoff's Rules & Networks	Kirchhoff's rules (junction rule and loop rule), applications to complex electrical circuits and loop analysis.	Junction rule (conservation of charge), loop rule (conservation of energy), systematic sign conventions across batteries and resistors.
Wheatstone Bridge & Measuring Instruments	Wheatstone bridge, balanced condition, sensitivity, meter bridge principle and applications.	Null deflection condition $\frac{P}{Q} = \frac{R}{S}$, conjugate arms invariance, meter bridge unknown resistance $X = \frac{R(100-l)}{l}$, end errors.

EXAM PATTERN

CUET UG Exam Architecture: In the CUET UG Physics examination, Unit 2 (Current Electricity) is one of the highest-yield topics, consistently contributing 5 to 6 MCQs out of 50. Questions range from straightforward numerical problems on cell combinations and drift velocity to conceptual questions on temperature dependence of resistivity and circuit reductions using Kirchhoff's rules.

Electric Current & Microscopic Conduction

Electric current quantifies the rate of flow of electric charge across a cross-sectional area of a conductor.

Electric Current & Current Density

If a net charge ΔQ crosses a normal cross-section in time interval Δt, the average current is:

$$I_{\text{avg}} = \frac{\Delta Q}{\Delta t}$$

The instantaneous electric current is the time derivative of charge:

$$I = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt} \quad [\text{SI unit: Ampere (A)} = \text{C/s}]$$

Electric current is a **scalar quantity** because it obeys ordinary laws of scalar algebra rather than vector addition, even though an arrow indicates the direction of positive charge flow.

CURRENT DENSITY VECTOR (J)

Current density (J) is a microscopic **vector quantity** defined as the electric current per unit cross-sectional area perpendicular to the direction of current flow:

$$(J) = \frac{dI}{dA_{\perp}}\hat{n} \implies I = \int (J) \cdot d(A) \quad [\text{SI unit: A/m}^2]$$

The direction of (J) aligns with the direction of motion of positive charge carriers (or opposite to the drift of electrons).

Microscopic Mechanism of Conduction: Drift Velocity & Mobility

Inside a metallic conductor at room temperature ($T \approx 300$ K):

- Free conduction electrons undergo random thermal motion with speeds $v_{\text{th}} \approx 10^5$ to 10^6 m/s. Because these thermal velocities are isotropic, the average velocity in any arbitrary direction is strictly zero:

$$(u)_{\text{avg}} = \frac{1}{N}\sum_{i=1}^{N}(u)_i = (0)$$

  Consequently, random thermal agitation generates zero net electric current across any cross-section.
- **Relaxation Time (τ):** The average time interval between two successive collisions of a conduction electron with vibrating metallic lattice ions. For metals at room temperature, $\tau \approx 10^{-14}$ s.
- **Action of an Applied Electric Field:** When an external potential difference establishes an electric field (E) within the conductor, each electron experiences an electrostatic force $(F) = -e(E)$, acquiring an acceleration:

$$(a) = -\frac{e(E)}{m}$$

- **Drift Velocity ($(v)_d$):** The average velocity with which conduction electrons drift opposite to the applied electric field under constant acceleration interrupted by collisions:

$$(v)_d = (u)_{\text{avg}} + (a)\tau = -\frac{e(E)\tau}{m}$$

  The magnitude of the drift velocity is:

$$v_d = \frac{eE\tau}{m} \quad [\text{typical magnitude: } \approx 10^{-4} \text{ m/s} = 0.1 \text{ mm/s}]$$

CRUCIAL CONCEPTUAL DISTINCTION: SIGNAL SPEED VS DRIFT SPEED

Although the physical drift speed of electrons is sluggish (≈ 0.1 mm/s), an electric bulb turns on almost instantaneously ($\approx 10^{-9}$ s) when switched on. This is because the electric field propagates through the circuit as an electromagnetic wave at nearly the speed of light ($c \approx 3 \times 10^8$ m/s), setting free electrons into drift throughout the conductor simultaneously.

Relation Between Electric Current and Drift Velocity

Consider a conductor of uniform cross-sectional area A containing n free conduction electrons per unit volume (number density). In a time interval Δt, electrons travel an average distance $\Delta x = v_d \Delta t$.

- The total volume traversed is $\Delta V = A\Delta x = Av_d\Delta t$.
- Total number of electrons within this volume: $N = n\Delta V = nAv_d\Delta t$.
- Total charge flowing across the cross-section: $\Delta Q = Ne = neAv_d\Delta t$.
- Therefore, the steady electric current is:

$$I = \frac{\Delta Q}{\Delta t} = neAv_d$$

- Dividing by cross-sectional area A, the current density magnitude is:

$$J = \frac{I}{A} = nev_d$$

MOBILITY (M)

Mobility μ is defined as the magnitude of drift velocity acquired per unit applied electric field:

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m} \quad [\text{SI unit: } \mathrm{m^2 \cdot V^{-1} \cdot s^{-1}}]$$

Mobility is **strictly positive**. Total electric current in a conductor containing multiple charge carrier species (electrons and holes, as in semiconductors or electrolytes) is:

$$I = eA\left(n_e v_{d,e} + n_h v_{d,h}\right) = eAE(n_e\mu_e + n_h\mu_h)$$

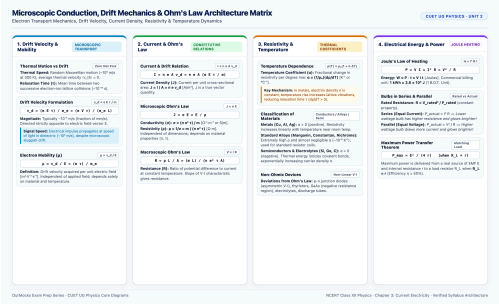


Figure 1: Microscopic Conduction Dynamics: Random thermal agitation, electric field induced drift velocity ($v_d = eE\frac{\tau}{m}$), and current density relations.

WORKED EXAMPLE 1: DRIFT VELOCITY CALCULATION

Problem: A copper wire of cross-sectional area 1.0×10^{-7} m^2 carries a steady current of 1.5 A. The conduction electron density of copper is 8.5×10^{28} m^{-3}. Calculate the drift speed of electrons.

Solution:

1. Formula: $I = neAv_d \Longrightarrow v_d = \frac{I}{neA}$.
2. Substitute parameters:

$$v_d = \frac{1.5}{(8.5 \times 10^{28}) \times (1.6 \times 10^{-19}) \times (1.0 \times 10^{-7})}$$

3. Multiply denominator terms:

$$neA = (8.5 \times 1.6) \times 10^{28-19-7} = 13.6 \times 10^2 = 1360 \text{ C/m}$$

4. Evaluate:

$$v_d = \frac{1.5}{1360} \approx 1.10 \times 10^{-3} \text{ m/s} = 1.10 \text{ mm/s}$$

Ohm's Law, Resistivity & Temperature Dynamics

Ohm's Law (Macroscopic & Microscopic Forms)

Ohm's law states that the steady current I flowing through a metallic conductor is directly proportional to the potential difference V maintained across its terminals, provided physical conditions (such as temperature, pressure, and mechanical strain) remain strictly constant:

$$V \propto I \implies V = IR$$

where R is the electrical resistance of the conductor (SI unit: Ohm (Ω) = V/A).

Derivation of Microscopic Ohm's Law & Resistivity

Using the fundamental drift velocity relation $v_d = \frac{eE\tau}{m}$ in $J = nev_d$:

$$J = ne\left(\frac{eE\tau}{m}\right) = \left(\frac{ne^2\tau}{m}\right)E$$

Defining electrical conductivity σ and electrical resistivity ρ:

$$\sigma = \frac{ne^2\tau}{m} \quad \text{and} \quad \rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$$

This yields the **microscopic (vector) form of Ohm's Law**:

$$(J) = \sigma(E) \quad \text{or} \quad (E) = \rho(J)$$

Resistance of a Conductor & Dependence on Geometry

For a conductor of uniform length L and cross-sectional area A with potential difference $V = EL$ and current $I = JA$:

$$V = EL = (\rho J)L = \rho\left(\frac{I}{A}\right)L = \left(\rho\frac{L}{A}\right)I$$

Comparing with $V = IR$ gives:

$$R = \rho\frac{L}{A} = \left(\frac{m}{ne^2\tau}\right)\frac{L}{A}$$

STRETCHING AND DRAWING A WIRE (VOLUME INVARIANCE)

When a wire of initial length L_1 and resistance R_1 is stretched uniformly to length $L_2 = kL_1$:

- Since the volume of the wire remains invariant ($V = A_1L_1 = A_2L_2$), cross-sectional area scales as $A_2 = \frac{A_1}{k}$.
- New resistance: $R_2 = \rho\frac{L_2}{A_2} = \rho\frac{kL_1}{\frac{A_1}{k}} = k^2\left(\rho\frac{L_1}{A_1}\right) = k^2R_1$.

- **Percentage change rule for small elongations ($\Delta\frac{L}{L} \leq 5\%$):**
$$\frac{\Delta R}{R} \approx 2\left(\frac{\Delta L}{L}\right) = -2\left(\frac{\Delta A}{A}\right) = -4\left(\frac{\Delta r}{r}\right)$$

Temperature Dependence of Resistance and Resistivity

The electrical resistivity of materials varies with temperature according to the relation:
$$\rho(T) = \rho_0[1 + \alpha(T - T_0)] \quad \Longrightarrow \quad R(T) = R_0[1 + \alpha(T - T_0)]$$
where α is the **temperature coefficient of resistance** ($^\circ C^{-1}$ or K^{-1}):
$$\alpha = \frac{1}{R_0}\frac{dR}{dT}$$

Material Class	Sign & Magnitude of α	Microscopic Physical Origin
Metallic Conductors (Cu, Al, Ag)	Positive ($\alpha > 0$), $\approx 4 \times 10^{-3}$ K^{-1}	Free electron density n is fixed; lattice vibrations increase with T, reducing collision relaxation time τ ($\rho \propto \frac{1}{\tau}$).
Alloys (Manganin, Constantan, Nichrome)	Positive, extremely small ($\alpha \approx 10^{-5}$ K^{-1})	High residual impurity scattering causes high base resistivity; nearly temperature-independent, ideal for standard resistance boxes.
Semiconductors & Carbon (Si, Ge)	Negative ($\alpha < 0$)	Thermal excitation rapidly elevates electron-hole carrier density $n(T) \propto e^{-\frac{E_g}{2k_BT}}$, dominating over minor decreases in τ.
Superconductors (Hg, YBCO)	Discontinuous drop to zero at T_c	Below critical temperature T_c, resistivity drops abruptly to identically zero ($\rho = 0$), expelling magnetic fields (Meissner effect).

Ohmic vs Non-Ohmic Devices

NON-OHMIC CONDUCTORS

Devices that do not satisfy linear proportionality between V and I are termed **non-ohmic**.

- **p-n Junction Diode:** Highly asymmetric $V - I$ curve; forward bias exhibits exponential current rise, reverse bias exhibits minute saturation current followed by breakdown.
- **Thyristor (SCR):** Shows regions of negative differential resistance ($d\frac{V}{d}I < 0$).
- **Electrolytes & Vacuum Tubes:** Show curved $V - I$ characteristics governed by child's law or ion mobility limits.
- **Dynamic Resistance:** Defined for non-ohmic devices at an operating point: $r_d = \frac{\Delta V}{\Delta I}$.

WORKED EXAMPLE 2: WIRE STRETCHING

Problem: A cylindrical copper wire of resistance 4.0Ω is drawn through a die such that its length increases by 25%. Assuming constant density and volume, find its new resistance.

Solution:

1. Initial resistance $R_1 = 4.0\Omega$.

2. Final length $L_2 = 1.25L_1 = \frac{5}{4}L_1$.
3. By volume conservation ($A_1L_1 = A_2L_2$), the new cross-sectional area is $A_2 = \frac{4}{5}A_1$.
4. Final resistance:

$$R_2 = \rho\frac{L_2}{A_2} = \rho\left(\frac{\frac{5}{4}L_1}{\frac{4}{5}A_1}\right) = \left(\frac{5}{4}\right)^2\left(\rho\frac{L_1}{A_1}\right) = \frac{25}{16}R_1$$

5. Substitute $R_1 = 4.0\Omega$:

$$R_2 = \frac{25}{16} \times 4.0 = \frac{25}{4} = 6.25\Omega$$

Electrical Energy, Joule's Law & Maximum Power

When an electric charge ΔQ moves across a potential difference V, the work done by the electrostatic field is $\Delta W = V\Delta Q = VI\Delta t$. In a purely resistive conductor, this electrical energy is dissipated entirely as thermal energy.

Joule's Law of Heating & Electrical Power

The rate of electrical energy dissipation in a resistance R carrying current I under terminal voltage V is:

$$P = \frac{dW}{dt} = VI = I^2R = \frac{V^2}{R} \quad \text{[SI unit: Watt (W) = J/s]}$$

Total heat energy liberated in time interval t is given by Joule's Law:

$$H = I^2Rt = VIt = \left(\frac{V^2}{R}\right)t \quad \text{[SI unit: Joule (J)]}$$

- **Commercial Unit of Electrical Energy:** 1 kilowatt-hour (kWh) = 1 Board of Trade Unit = $1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6$ J.

Bulb Ratings, Series and Parallel Illuminations

Incandescent lamps are labelled with rated power P_r and rated voltage V_r (typically 220 V).

- The internal resistance of the filament is fixed by its manufactured geometry and material:

$$R = \frac{V_r^2}{P_r}$$

Therefore, **higher wattage corresponds to lower filament resistance** ($R \propto \frac{1}{P_r}$).

SERIES VS PARALLEL BULB COMBINATIONS

1. **In Series Combination (I is identical for all bulbs):**

$$P_{\text{consumed}} = I^2R \propto R \propto \frac{1}{P_r}$$

The bulb with the **lowest rated wattage** has the highest resistance, dissipates the most power, and glows with the **greatest brightness**.

2. **In Parallel Combination (V is identical across all bulbs):**

$$P_{\text{consumed}} = \frac{V^2}{R} \propto \frac{1}{R} \propto P_r$$

The bulb with the **highest rated wattage** has the lowest resistance, draws the most current, dissipates the most power, and glows with the **greatest brightness**.

Maximum Power Transfer Theorem

Consider a real DC source of fixed EMF E and internal resistance r connected to an external variable load resistance R_L:

- Current delivered to the load: $I = \frac{E}{R_L + r}$.
- Power consumed in the load resistance:

$$P_L = I^2 R_L = E^2 \frac{R_L}{(R_L + r)^2}$$

- To find the condition for maximum power transfer, differentiate P_L with respect to R_L and set to zero:

$$\frac{dP_L}{dR_L} = E^2 \frac{\left[(R_L + r)^2 - 2R_L(R_L + r)\right]}{(R_L + r)^4} = 0 \Longrightarrow (R_L + r) - 2R_L = 0 \Longrightarrow R_L = r$$

- Under this matching condition ($R_L = r$), the **maximum power dissipated** in the load is:

$$P_{L,\max} = E^2 \frac{r}{(r + r)^2} = \frac{E^2}{4r}$$

- **Efficiency at Maximum Power:** The internal source resistance dissipates equal power $P_r = I^2 r = \frac{E^2}{4r}$. The efficiency of the circuit is only $\eta = \frac{P_L}{P_L + P_r} = 50\%$.

WORKED EXAMPLE 3: SERIES VS PARALLEL BULBS

Problem: Two electric bulbs rated 220 V, 40 W and 220 V, 100 W are connected in series across a 220 V AC mains supply. Which bulb glows brighter, and what is their total power consumption?

Solution:

1. Calculate individual resistances:

$$R_1 = \frac{V_r^2}{P_1} = \frac{220^2}{40} = 1210\Omega, \quad R_2 = \frac{V_r^2}{P_2} = \frac{220^2}{100} = 484\Omega$$

2. Total series resistance: $R_{\text{total}} = R_1 + R_2 = 1210 + 484 = 1694\Omega$.
3. Total power drawn from 220 V:

$$P_{\text{total}} = \frac{V^2}{R_{\text{total}}} = \frac{220^2}{1694} = \frac{48400}{1694} \approx 28.57 \text{ W}$$

4. Actual power consumed individually:
   - $P_{1'} = I^2 R_1 \propto R_1 = 1210\Omega$.
   - $P_{2'} = I^2 R_2 \propto R_2 = 484\Omega$.
5. Since $R_1 > R_2$, the 40 W bulb dissipates more power and glows **much brighter**.

Electromotive Force, Internal Resistance & Combination of Cells

An electrochemical cell maintains a continuous electric current by converting chemical energy into electrical potential energy through redox reactions.

EMF (E), Terminal Voltage (V) and Internal Resistance (r)

- **Electromotive Force (E):** The potential difference between the terminals of a cell when no current is drawn (open-circuit condition, $I = 0$). It represents the non-electrostatic work done per unit positive charge by internal chemical forces to move it from the cathode to the anode.
- **Internal Resistance (r):** The finite opposition offered by the electrolyte and internal electrodes to the flow of ionic current within the cell. Factors affecting r:
  - Directly proportional to the separation between electrodes (d).
  - Inversely proportional to the submerged surface area of electrodes (A).
  - Directly proportional to electrolyte concentration (C) and inversely proportional to temperature (T).

TERMINAL POTENTIAL DIFFERENCE EQUATIONS

1. **Discharging Condition (Cell supplies current to an external circuit):**
$$V = E - Ir \implies I = \frac{E - V}{r} = \frac{E}{R + r}$$
Here, terminal potential difference is strictly less than EMF ($V < E$).
2. **Charging Condition (External source forces current into positive terminal):**
$$V = E + Ir$$
Here, terminal potential difference strictly exceeds EMF ($V > E$).
3. **Open Circuit Condition ($I = 0$):** $V = E$.
4. **Short Circuit Condition ($R = 0$):** $V = 0$ and current hits maximum: $I_{\text{sc}} = \frac{E}{r}$.
5. **Internal Resistance Formula:** Expressed in terms of load R, EMF E, and terminal voltage V:
$$r = \left(\frac{E - V}{V}\right) R = \left(\frac{E}{V} - 1\right) R$$

Combination of Identical Cells

Configuration	Equivalent EMF & Resistance	Maximum Current Condition
Series Combination (n cells)	$E_{\text{eq}} = nE$ $r_{\text{eq}} = nr$ $I = \frac{nE}{R+nr}$	▸ If $R \gg nr$: $I \approx n\left(\frac{E}{R}\right)$ (n times single cell current; **optimal**). ▸ If $R \ll nr$: $I \approx \frac{E}{r}$ (no benefit).
Parallel Combination (m cells)	$E_{\text{eq}} = E$ $r_{\text{eq}} = \frac{r}{m}$ $I = \frac{E}{R+\frac{r}{m}} = \frac{mE}{mR+r}$	▸ If $R \ll \frac{r}{m}$: $I \approx m\left(\frac{E}{r}\right)$ (m times single cell current; **optimal**). ▸ If $R \gg \frac{r}{m}$: $I \approx \frac{E}{R}$ (no benefit).
Mixed Grouping (n series in m rows)	$N = nm$ total cells $E_{\text{eq}} = nE$ $r_{\text{eq}} = \frac{nr}{m}$ $I = \frac{nE}{R+n\frac{r}{m}} = \frac{mnE}{mR+nr}$	Current is maximized when load resistance equals total internal resistance: $R = \frac{nr}{m} \implies mR = nr$

Configuration	Equivalent EMF & Resistance	Maximum Current Condition
		Maximum current: $I_{\max} = \frac{nE}{2R} = \frac{mE}{2r}$.

Combination of Unequal Cells in Parallel

Consider two non-identical cells of EMFs E_1, E_2 and internal resistances r_1, r_2 connected in parallel across terminals A and B:

$$E_{\text{eq}} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \quad \text{and} \quad \frac{1}{r_{\text{eq}}} = \frac{1}{r_1} + \frac{1}{r_2} \Longrightarrow r_{\text{eq}} = \frac{r_1 r_2}{r_1 + r_2}$$

If cell 2 is reversed in polarity: $E_{\text{eq}} = \frac{E_1 r_2 - E_2 r_1}{r_1 + r_2}$.

WORKED EXAMPLE 4: UNEQUAL CELLS IN PARALLEL

Problem: Two batteries of EMF 6 V and 12 V with internal resistances 1Ω and 2Ω are connected in parallel with like terminals joined together. Find the equivalent EMF and internal resistance of the combination.

Solution:

1. Equivalent internal resistance:
$$r_{\text{eq}} = \frac{r_1 r_2}{r_1 + r_2} = \frac{1 \times 2}{1 + 2} = \frac{2}{3}\Omega \approx 0.67\Omega$$
2. Equivalent EMF:
$$E_{\text{eq}} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{6}{1} + \frac{12}{2}}{\frac{1}{1} + \frac{1}{2}} = \frac{6 + 6}{\frac{3}{2}} = \frac{12}{1.5} = 8.0 \text{ V}$$
3. The equivalent battery acts as an 8.0 V EMF source with internal resistance $\frac{2}{3}\Omega$.

Kirchhoff's Rules & Network Analysis

Complex multi-loop electrical networks containing multiple EMF sources and cross-connected resistors cannot be simplified by simple series-parallel reduction. Gustav Kirchhoff formulated two fundamental conservation laws to systematically solve arbitrary circuits.

Kirchhoff's First Rule: The Junction Rule (KCL)

At any circuit junction (node) where three or more conductors meet, the algebraic sum of currents is identically zero:

$$\sum_{k=1}^{N} I_k = 0 \quad \Longrightarrow \quad \sum I_{\text{inflow}} = \sum I_{\text{outflow}}$$

- **Physical Foundation:** Kirchhoff's Junction Rule is a direct manifestation of the **Law of Conservation of Electric Charge**. Since charge cannot accumulate or deplete at an infinitesimally small junction point in a steady-state circuit ($d\frac{Q}{d}t = 0$), incoming charge per unit time must equal outgoing charge per unit time.

Kirchhoff's Second Rule: The Loop Rule (KVL)

Around any closed conducting loop in an electrical network, the algebraic sum of changes in potential is identically zero:

$$\sum \Delta V = 0 \implies \sum E = \sum IR$$

- **Physical Foundation:** Kirchhoff's Loop Rule is a direct consequence of the **Law of Conservation of Energy** in a conservative electrostatic field. When a unit positive charge travels in a complete closed trajectory back to its starting point, the net work done by conservative electrostatic and source forces is identically zero ($\oint (E) \cdot d(l) = 0$).

RIGOROUS SIGN CONVENTIONS FOR LOOP TRAVERSAL

When traversing a closed loop in a chosen clockwise or counter-clockwise sense:

1. **Traversing a Resistor:**
   - Moving in the **same direction as current I**: Potential drops ($\Delta V = -IR$).
   - Moving in the **direction opposite to current I**: Potential rises ($\Delta V = +IR$).
2. **Traversing an EMF Source:**
   - Moving from **negative to positive terminal** (inside source): Potential rises ($\Delta V = +E$), independent of current direction.
   - Moving from **positive to negative terminal** (inside source): Potential drops ($\Delta V = -E$), independent of current direction.

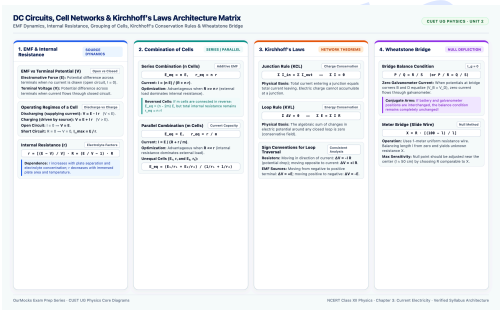


Figure 2: Circuit Analysis Architecture: Kirchhoff's junction & loop laws, sign traversal rules, and Wheatstone bridge null deflection topology.

WORKED EXAMPLE 5: TWO-LOOP KIRCHHOFF NETWORK

Problem: In a two-loop circuit, loop 1 contains source $E_1 = 10$ V, internal resistance $r_1 = 1\Omega$, and branch resistance $R_1 = 2\Omega$. Loop 2 contains source $E_2 = 4$ V, internal resistance $r_2 = 1\Omega$, and branch resistance $R_2 = 4\Omega$. Both feed a central common load resistor $R_3 = 3\Omega$. Find the current I_3 through the load resistor.

Solution:

1. Let I_1 flow from E_1 and I_2 from E_2. By KCL at the upper junction, load current is $I_3 = I_1 + I_2$.
2. KVL around loop 1 (containing E_1 and R_3):
   $E_1 - I_1(r_1 + R_1) - (I_1 + I_2)R_3 = 0 \implies 10 - 3I_1 - 3(I_1 + I_2) = 0 \implies 6I_1 + 3I_2 = 10 \quad$ (Eq 1)
3. KVL around loop 2 (containing E_2 and R_3):
   $E_2 - I_2(r_2 + R_2) - (I_1 + I_2)R_3 = 0 \implies 4 - 5I_2 - 3(I_1 + I_2) = 0 \implies 3I_1 + 8I_2 = 4 \quad$ (Eq 2)
4. Multiply Eq 2 by 2 and subtract Eq 1:

$$2(3I_1 + 8I_2) - (6I_1 + 3I_2) = 8 - 10 \Longrightarrow 13I_2 = -2 \Longrightarrow I_2 = -\frac{2}{13} \text{ A}$$

5. Solve for I_1 from Eq 1:

$$6I_1 = 10 - 3\left(-\frac{2}{13}\right) = 10 + \frac{6}{13} = \frac{136}{13} \Longrightarrow I_1 = \frac{136}{78} = \frac{68}{39} \text{ A}$$

6. Current through load resistor R_3:

$$I_3 = I_1 + I_2 = \frac{68}{39} - \frac{6}{39} = \frac{62}{39} \approx 1.59 \text{ A}$$

Wheatstone Bridge & Measuring Instruments

Wheatstone Bridge Architecture & Null Deflection

The Wheatstone bridge consists of four resistors P, Q, R, S arranged along the four arms of a quadrilateral $ABCD$. A galvanometer G is connected across bridge arm BD, and a DC EMF source is connected across conjugate terminals AC.

BALANCED WHEATSTONE BRIDGE CONDITION

When the resistances are adjusted such that **no current flows through the galvanometer** ($I_g = 0$), points B and D reside at identical potentials ($V_B = V_D$). The bridge is balanced if and only if:

$$\frac{P}{Q} = \frac{R}{S}$$

- **Proof using KVL:**
  - Loop $ABDA$ with $I_g = 0$: $I_1P - I_2R = 0 \Longrightarrow I_1P = I_2R$.
  - Loop $BCDB$ with $I_g = 0$: $I_1Q - I_2S = 0 \Longrightarrow I_1Q = I_2S$.
  - Dividing the two equations: $\frac{I_1P}{I_1Q} = \frac{I_2R}{I_2S} \Longrightarrow \frac{P}{Q} = \frac{R}{S}$.

CRUCIAL PROPERTIES OF BALANCED WHEATSTONE BRIDGE

1. **Conjugate Arms Property:** Interchanging the positions of the battery and the galvanometer does **not** alter the balance condition.
2. **Maximum Sensitivity:** The Wheatstone bridge achieves highest measurement precision when all four arms have resistances of comparable magnitude ($P \approx Q \approx R \approx S$).
3. **Equivalent Resistance:** When balanced, the central galvanometer branch carries no current and can be omitted:

$$R_{\text{eq}} = \frac{(P+Q)(R+S)}{P+Q+R+S}$$

Meter Bridge (Slide Wire Bridge)

The meter bridge operates on the principle of the balanced Wheatstone bridge. It consists of a uniform manganin or constantan wire of length 100 cm (1 m) stretched over a graduated meter scale:

- A known standard resistance R is placed in the left gap, and an unknown resistance $S = X$ is placed in the right gap.

- The jockey is moved along the wire until the galvanometer registers null deflection ($I_g = 0$) at balance length l from the left end (A).
- The resistance of wire segment AJ is $P = r_0 l$, and segment JC is $Q = r_0(100 - l)$, where r_0 is resistance per unit length.
- By Wheatstone bridge balance:

$$\frac{P}{Q} = \frac{R}{X} \implies \frac{r_0 l}{r_0(100-l)} = \frac{R}{X} \implies X = R\left(\frac{100-l}{l}\right)$$

EXAM PATTERN

Minimizing Experimental Errors in Meter Bridge:

- **End Resistance Error:** Caused by finite contact resistance at the copper strip joints and soldering points. It is eliminated by repeating the measurement after interchanging R and X in the two gaps and averaging the results.
- **Optimal Balance Point:** Sensitivity is maximized and relative measurement error is minimized ($d\frac{X}{X} \to$ minimum) when the null point falls near the exact center of the wire ($l \approx 50$ cm).

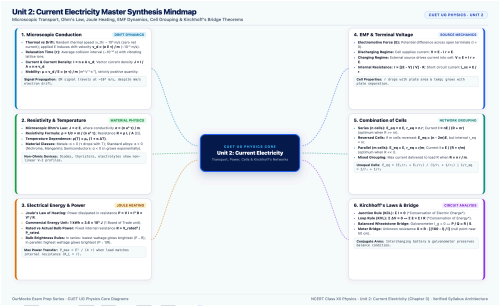


Figure 3: Master Synthesis Mindmap: Microscopic conduction, Ohm's law, Joule heating, cell grouping, and bridge analysis.

MASTER CURRENT ELECTRICITY REVISION FORMULA CHEATSHEET

- **Current & Drift Velocity:** $I = neAv_d$ | $v_d = \frac{eE\tau}{m}$ | $(J) = ne(v)_d = \sigma(E)$ | $\mu = \frac{v_d}{E} = \frac{e\tau}{m}$
- **Resistance & Resistivity:** $R = \rho\frac{L}{A}$ | $\rho = \frac{m}{ne^2\tau}$ | Stretched k times: $R' = k^2R$ | $\Delta\frac{R}{R} \approx 2(\Delta\frac{L}{L})$
- **Temperature Dependence:** $R(T) = R_0(1 + \alpha\Delta T)$ | Metals $\alpha > 0$, Semiconductors $\alpha < 0$
- **Heating & Power:** $P = VI = I^2R = \frac{V^2}{R}$ | Bulbs in series: $P_{\text{glow}} \propto R$; Bulbs in parallel: $P_{\text{glow}} \propto \frac{1}{R}$
- **Max Power Transfer:** Load matches internal resistance: $R_L = r \implies P_{\max} = \frac{E^2}{4r}$ with 50% efficiency
- **Cells:** Discharging $V = E - Ir$ | Charging $V = E + Ir$ | Internal resistance $r = \left(\frac{E-V}{V}\right)R$
- **Cell Combinations:** Series nE, nr | Parallel $E, \frac{r}{m}$ | Unequal parallel: $E_{\text{eq}} = \sum \frac{\frac{E_i}{r_i}}{\sum}\left(\frac{1}{r_i}\right)$
- **Kirchhoff's Laws:** KCL $\sum I_{\text{in}} = \sum I_{\text{out}}$ (Charge) | KVL $\sum \Delta V = 0$ (Energy)
- **Wheatstone Bridge:** Balanced condition $\frac{P}{Q} = \frac{R}{S}$ | Meter bridge: $X = R\frac{100-l}{l}$

CUET UG Exam-Style Practice Questions

Q1. A copper wire of resistance R is stretched uniformly so that its length increases by 0.2%. The percentage change in its electrical resistance is approximately:

(A) $+0.2\%$ (B) $+0.4\%$
(C) -0.4% (D) $+0.1\%$

Sol. Q1. By volume invariance, $R \propto L^2$. For small fractional changes ($\leq 5\%$), $\frac{\Delta R}{R} \approx 2\frac{\Delta L}{L} = 2(+0.2\%) = +0.4\%$. Resistance increases by 0.4%.

Answer: (B)

Q2. When a steady current passes through a cylindrical metallic conductor of non-uniform cross-section, which of the following quantities remains constant along the length?

(A) Current density (B) Electric field
(C) Drift speed of electrons (D) Electric current

Sol. Q2. By conservation of charge, electric current I is strictly constant across every cross-section. Current density $J = \frac{I}{A}$, electric field $E = \frac{J}{\sigma}$, and drift velocity $v_d = \frac{I}{neA}$ all vary inversely with cross-sectional area A.

Answer: (D)

Q3. A potential difference V is applied across a copper wire of length L and diameter d. If V is doubled and d is halved while L remains constant, the drift velocity of electrons will:

(A) Double (B) Halve
(C) Quadruple (D) Remain unchanged

Sol. Q3. Drift speed $v_d = \frac{eE\tau}{m} = \frac{eV\tau}{mL}$. Drift velocity depends strictly on electric field $E = \frac{V}{L}$ and is completely independent of wire diameter d. Doubling V doubles v_d.

Answer: (A)

Q4. A carbon resistor has colored bands in the sequence: Yellow, Violet, Brown, and Gold. Its resistance value according to the standard color code is:

(A) $(47 \times 10^1 \pm 5\%)\Omega$ (B) $(47 \times 10^2 \pm 10\%)\Omega$
(C) $(57 \times 10^1 \pm 5\%)\Omega$ (D) $(47 \times 10^0 \pm 20\%)\Omega$

Sol. Q4. Color code: Yellow = 4, Violet = 7, Brown = 10^1 multiplier, Gold = $\pm 5\%$ tolerance. Thus, $R = (47 \times 10^1 \pm 5\%)\Omega = (470 \pm 5\%)\Omega$.

Answer: (A)

Q5. The temperature coefficient of resistance is positive for:

(A) Germanium (B) Silicon
(C) Copper (D) Carbon

Sol. Q5. Metals like copper have positive temperature coefficients ($\alpha > 0$) because increased lattice vibrations decrease relaxation time τ. Silicon, germanium, and carbon have negative coefficients ($\alpha < 0$) due to covalent bond breakdown.

Answer: (C)

Q6. A cell of EMF E and internal resistance r is connected across a variable external load resistor R. The maximum power dissipated in the external resistor is:

(A) $\frac{E^2}{2r}$ (B) $\frac{E^2}{4r}$
(C) $\frac{E^2}{r}$ (D) $\frac{2E^2}{r}$

Sol. Q6. By the maximum power transfer theorem, power dissipated in external load is maximized when $R = r$. Maximum power $P_{\max} = E^2 \frac{R}{(R+r)^2} = E^2 \frac{r}{(2r)^2} = \frac{E^2}{4r}$.

Answer: (B)

Q7. Two electric bulbs rated 25 W – 220 V and 100 W – 220 V are connected in series across a 440 V supply. Which bulb will fuse?

(A) 100 W bulb (B) 25 W bulb
(C) Both bulbs (D) Neither bulb

Sol. Q7. $R_1 = \frac{220^2}{25} = 1936\Omega$, $R_2 = \frac{220^2}{100} = 484\Omega$. Total $R = 2420\Omega$. Current from 440 V is $I = \frac{440}{2420} = \frac{2}{11}$ A. Potential across 25 W bulb is $V_1 = IR_1 = \left(\frac{2}{11}\right)(1936) = 352$ V. Since 352 V $>$ 220 V, the 25 W bulb fuses instantly.

Answer: (B)

Q8. Kirchhoff's first and second laws of electrical circuits are based on the conservation of:

(A) Energy and electric charge respectively (B) Electric charge and energy respectively
(C) Electric charge and momentum respectively (D) Energy and angular momentum respectively

Sol. Q8. Kirchhoff's first rule (junction rule, $\sum I = 0$) embodies conservation of electric charge. Kirchhoff's second rule (loop rule, $\sum \Delta V = 0$) embodies conservation of energy in a conservative field.

Answer: (B)

Q9. A battery of EMF 10 V and internal resistance 3Ω is connected to a resistor. If the current in the circuit is 0.5 A, the terminal voltage of the battery is:

(A) 8.5 V (B) 10.0 V
(C) 1.5 V (D) 7.0 V

Sol. Q9. During discharge, terminal potential difference is $V = E - Ir = 10 - (0.5 \times 3) = 10 - 1.5 = 8.5$ V.

Answer: (A)

Q10. In a meter bridge experiment, the null point is obtained at 40 cm from the left end when a resistance of 6Ω is in the left gap. The unknown resistance in the right gap is:

(A) 4Ω (B) 9Ω
(C) 6Ω (D) 12Ω

Sol. Q10. Using meter bridge balance condition: $\frac{R}{X} = \frac{l}{100-l} \Longrightarrow \frac{6}{X} = \frac{40}{100-40} = \frac{40}{60} = \frac{2}{3} \Longrightarrow X = \frac{6\times 3}{2} = 9\Omega$.

Answer: (B)

Q11. Two identical cells, each of EMF 1.5 V and internal resistance 0.5Ω, are connected in parallel. The equivalent EMF and equivalent internal resistance of the combination are:

(A) 3.0 V, 1.0Ω (B) 1.5 V, 0.25Ω
(C) 1.5 V, 1.0Ω (D) 3.0 V, 0.25Ω

Sol. Q11. In parallel combination of identical cells, equivalent EMF equals single cell EMF: $E_{\text{eq}} = E = 1.5$ V. Equivalent internal resistance is $r_{\text{eq}} = \frac{r}{m} = \frac{0.5}{2} = 0.25\Omega$.

Answer: (B)

Q12. In a Wheatstone bridge network $P = 2\Omega, Q = 4\Omega, R = 3\Omega, S = 6\Omega$. A galvanometer of resistance 10Ω is connected across the bridge. The current through the galvanometer is:

(A) 1 A (B) 0.5 A
(C) 0 A (D) 2 A

Sol. Q12. Check balance condition: $\frac{P}{Q} = \frac{2}{4} = \frac{1}{2}$ and $\frac{R}{S} = \frac{3}{6} = \frac{1}{2}$. Since $\frac{P}{Q} = \frac{R}{S}$, the bridge is perfectly balanced and zero current flows through the galvanometer ($I_g = 0$).

Answer: (C)

Q13. A wire of resistance 12Ω is bent in the form of a closed uniform circle. The effective resistance between two diametrically opposite points on the circle is:

(A) 6Ω (B) 3Ω
(C) 12Ω (D) 24Ω

Sol. Q13. Two diametrically opposite points divide the circle into two semicircular segments of equal resistance $R_1 = R_2 = \frac{12}{2} = 6\Omega$ connected in parallel: $R_{\text{eq}} = \frac{6\times6}{6+6} = 3\Omega$.

Answer: (B)

Q14. A heater coil is cut into two equal halves and only one half is connected across the same mains supply. The heat generated per second compared to the original coil will be:

(A) Halved (B) Doubled
(C) Quadrupled (D) Unchanged

Sol. Q14. Resistance of half coil is $R' = \frac{R}{2}$. Rate of heat generated at constant supply voltage V is $\frac{H}{t} = \frac{V^2}{R}$. With new resistance R', $P' = \frac{V^2}{\frac{R}{2}} = 2\left(\frac{V^2}{R}\right) = 2P$. Heat generation is doubled.

Answer: (B)

Q15. In a potentiometer or meter bridge circuit, balance point is obtained near the center (50 cm) of the slide wire primarily to:

(A) Maximize sensitivity and minimize percentage measurement error (B) Prevent heating of the slide wire
(C) Ensure the galvanometer does not burn out (D) Double the unknown resistance

Sol. Q15. The Wheatstone bridge is most sensitive when all four arms have comparable resistance values ($P \approx Q$). For a meter bridge wire, obtaining the null point near $l = 50$ cm minimizes fractional error $d\frac{X}{X}$ arising from meter rule reading uncertainties and end-resistance offsets.

Answer: (A)